Saptarshi Mandal

Fourth year undergraduate,
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EDUCATION

Indian Institute of Technology Kanpur
BS in Mathematics & Scientific Computing (CPI: 8.9 / 10.0) 2023 - 2027 (ongoing)
Minor in Theory of Computation
MS in Mathematics 2027 - 2028 (upcoming)

INTERESTS

Set Theory, Model Theory & Logic and its applications to other areas of mathematics
Geometric Measure Theory, Fourier & Harmonic Analysis, General Topology

PRESENTATIONS

"Hat Guessing with Proper Colorings", **Joint Mathematics Meetings 2026**, Washington DC, USA, January 2026.

PREPRINTS

"Hat Guessing with Proper Colorings", **Saptarshi Mandal**, Anurag Ramachandran, Michael Kreiger, Peter Bentley, 2025. [arXiv]

RESEARCH

Sticky Kakeya Conjecture | ICTS-TIFR Summer Research May 2026 - Jul 2026
Dr. Trishen Gunaratnam (TIFR Mumbai)
• Studied the *Wang-Zahl (2022)* proof of the sticky Kakeya conjecture in detail, learning geometric measure theory alongside; understood the full argument except the proof of 5.2
• Delivered a 20-minute presentation on the structural properties: coarse self-similarity, plaininess, graininess; wrote a detailed report expanding on several proofs from the paper.

Hat Guessing Number of Graphs | Polymath Jr., 2025 May 2025 - Jul 2025
Dr. Anurag Bishnoi (TU Delft)
• Proved that the hat guessing number of the complete graph K_n is $2n - 1$.
• Assisted in the proofs of some general bounds on the proper hat guessing number.

Abstract Harmonic Analysis | SURGE 2025, IIT Kanpur May 2025 - July 2025
Dr. Arijit Ganguly (IIT Kanpur)
• Studied basic Fourier theory for Hilbert Spaces and LCA Groups.
• Reconstructed proofs of Fubini's Theorem and the General Plancherel Theorem.

PROJECTS

Topics in set theory & model theory | Undergraduate Project Jul 2026 - present
Dr. Ashutosh Kumar (IIT Kanpur)
• Studying *Keisler*'s survey on ultraproducts to build a strong background in model theory.
• Working towards *Malliaris-Shelah (2014)* proof of $\mathfrak{p} = \mathfrak{t}$ over the coming months.

Topics in number theory | Undergraduate Project Jul 2026 - present
Dr. Somnath Jha (IIT Kanpur)
• Reading *Serre*'s *A Course in Arithmetic* with the goal of building a working knowledge of analytic number theory.

Function with a divergent Fourier Series | *course presentation*
Riesz-Thorin Interpolation Theorem | *course presentation*
Prime factorization of ideals in a Dedekind domain | *course presentation*

Type theory & Functional Programming May 2024 - July 2024
Stamatics Society, IIT Kanpur
• Explored fundamental type theory concepts, like type inference and polymorphism, algebraic data types, parametric polymorphism, monads and concurrency.
• Implemented the above in OCaml.

ACHIEVEMENTS

ICTS - **S.N. Bhatt Memorial Excellence Fellowship Program** 2026
Academic Excellence Award, IIT Kanpur (top 10% students) 2024
Rank 2848, JEE Advanced (out of 150,000 candidates) 2023
Rank 7449, JEE Mains (out of 1.4 million candidates) 2023
Rank 66, WBJEE (out of 120,000 candidates) 2023

COURSEWORK

Logic: Set Theory & Logic; Data Structures & Algorithms; Fundamentals of Computing; Topics in Model Theory*; Computability Theory*; Theory of Computation*

Analysis: Real Analysis, Complex Analysis, Measure Theory, Ordinary Differential Equations, Partial Differential Equations, Euclidean Fourier Analysis, Functional Analysis*

Algebra: Linear Algebra, Abstract Algebra, Field & Galois Theory, Algebraic Topology, Arithmetic on Quadratic Fields

Others: Several Variable Calculus & Differential Geometry, Probability Theory, Numerical Analysis & Scientific Computing I & II, Digital Electronics

SKILLS

Programming: C, C++, Python, Rust, Bash, Prolog, OCaml
Utilities: Git, Linux, Latex, MATLAB, SageMath

POSITIONS

Coordinator, Stamatics Society, IIT Kanpur Oct 2025 - present
General Executive, Stamatics Society, IIT Kanpur Nov 2024 - Oct 2025
Blog Writer, Stamatics Society, IIT Kanpur Nov 2024 - Oct 2025
Student Guide, Institute Counselling Service, IIT Kanpur Aug 2024 - present
Academic Mentor, Institute Counselling Service, IIT Kanpur Aug 2024 - present